

Shiping Gao

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RESEARCH INTERESTS

NLP, LLMs, RLHF, Reward Hacking, Fine-Grained Reward Modeling, and LLM Reasoning.

EDUCATION BACKGROUND

University of Michigan (UMich) Ann Arbor, MI, USA Sep 2026 – Jun 2030 (Expected)

- PhD. in Computer Science and Engineering, Advisor: Silviu Pitis.

Sun Yat-Sen University (SYSU, Project 985) Guangzhou, China Sep 2023 – Jun 2026

- MEng. in Computer Science and Technology, GPA: 89.84/100.

Lanzhou University (LZU, Project 985) Lanzhou, China Sep 2019 – Jul 2023

- B.Eng. in Computer Science and Technology, GPA: 90.17/100, Ranking: 8/232 (top 3.5%).
- Honors: Outstanding Graduate and Outstanding Graduation Thesis in 2023 (top 10%), Hui-Chun Chin & Tsung-Dao Lee Undergraduate Endowment (CURE) in 2022 (a competitive research funding for students from 6 top Chinese universities, selected as the only award-winning project from CS department with a Transformer Model for Low-resource Tibetan-Chinese Machine Translation), and First Prize Scholarship in 2021 (top 5%).

PAPERS

- Y Zhu, J Guo, S Gao, T Yang, X Quan. Token-Adaptive Entropy Regularization for On-Policy Distillation. Submitted to the 41st Annual AAAI Conference on Artificial Intelligence (AAAI2027 Submission)
- H Chen, T Yang, Y Zhu, S Gao, X Quan, T Yao. *Stabilizing Policy Optimization via Logits Convexity*. Submitted to the 40th Annual Conference on Neural Information Processing Systems (NeurIPS2026 Submission)
- H Li, R Ni, H Yang, S Gao, Y Liu, X Quan. *Adaptive Boundaries: Context-Aware Detection for Synthetic Text*. The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP2026)
- S Gao, H Chen, X Quan, Q Wang, L Huang. *Unleashing Implicit Rewards: Prefix-Value Learning for Distribution-Level Optimization*. 43rd International Conference on Machine Learning (ICML2026)
- H Chen, T Yang, S Gao, R Chen, X Quan, H Tian, T Yao. *Discriminative Policy Optimization for Token-Level Reward Models*. 42nd International Conference on Machine Learning (ICML2025 Poster)
- S Gao, F Wan, J Guo, X Quan, Q Wang. *Advantage-Guided Distillation for Preference Alignment in Small Language Models*. 13th International Conference on Learning Representations (ICLR2025 Spotlight Top 5.1%)
- J Liang, H Yang, S Gao, X Quan. *Edit-Wise Preference Optimization for Grammatical Error Correction*. Proceedings of the 31st International Conference on Computational Linguistics (COLING2025 Main)
- R Chen, J Liang, S Gao, F Wan, X Quan. *Self-Evolution Fine-Tuning for Policy Optimization*. Findings of the Association for Computational Linguistics (Findings of EMNLP 2024)
- S Gao, D He, Z Zhang, X Tang, Z Zhao. *A Novel Dynamic Interpolation Method based on both Temporal and Spatial Correlations*. Applied Intelligence. 2022. Springer (SCI JCR Q2)

RESEARCH EXPERIENCE

Unleashing Implicit Rewards: Prefix-Value Learning for Distribution-Level Optimization. Jul 2025 – Jan 2026

Internship, Supervised by Prof. Lifu Huang at University of California, Davis

- Developed an **Implicit Prefix-Value Reward Model (IPVRM)** in a *value-function style* with supervision on every prefix, resolving the training–inference inconsistency of prior implicit PRMs. The resulting IPVRM preserves the efficiency of prior IM-RMs while significantly improving the detection of erroneous reasoning steps in math tasks, enabling a 0.6B model to achieve performance comparable to other 7B PRMs.
- Pioneered the incorporation of **distribution-level temporal-difference advantages into RL**, enabling dense credit assignment for unsampled tokens. Combined with token-level GAE, this approach balanced long-horizon returns with efficiency, empirically demonstrating substantial improvements in RL convergence.

Discriminative Policy Optimization for Token-Level Reward Models (Q-RM)

Oct 2024 – Jan 2025

Research Team Member, Supervised by Prof. Xiaojun Quan at Sun Yat-Sen University

- Designed **Q-RM**, a novel token-level reward model that learns token-level rewards without fine-grained annotations by decoupling reward modeling from language modeling via **discriminative policy**, addressing instability in traditional token-level DPO-RM.
- Derived a token-level Q-function from a discriminative perspective, theoretically equivalent to the optimal Q-function; integrated Q-RM into **PPO/REINFORCE**, enabling direct computation of advantages from discriminative logits and reducing reliance on GAE.
- Demonstrated **4–6 point Pass@1 improvements** on GSM8K and MATH reasoning tasks; achieved **12× faster convergence than ORM** and **11× faster than step-level PRM**; also outperformed ORM and PRM baselines in reading comprehension and instruction-following tasks.

Advantage-Guided Distillation for Preference Alignment in Small Language Models

Jul 2024 – Oct 2024

Principal Investigator, Supervised by Prof. Xiaojun Quan at Sun Yat-Sen University

- Designed Dual-Constrained Knowledge Distillation (DCKD), which applies KL-divergence constraints on both preferred and dispreferred responses, enabling richer transfer of alignment knowledge.
- Introduced Advantage-Guided Distillation for Preference Alignment (ADPA), leveraging **log-probability margins** between aligned and unaligned teacher LLMs to provide distribution-level preference signals.
- Developed ADPA+, DCKD initialization followed by ADPA, achieving superior performance and stability.
- Demonstrated that ADPA/ADPA+ significantly improve alignment on MT-Bench, AlpacaEval, and the Open LLM Leaderboard, and small models trained with our methods on HelpSteer2 and DPO-MIX-7K matched or even surpassed large-model alignment baselines, while reducing sample complexity and stabilizing training.

Edit-Wise Preference Optimization for Grammatical Error Correction

Jul 2024 – Sep 2024

Research Team Member, Supervised by Prof. Xiaojun Quan at Sun Yat-Sen University

- Proposed **Edit-Wise Preference Optimization (EPO)** for grammatical error correction, which incorporates **edit-aware token weighting** by emphasizing tokens in edit spans and especially **pivot tokens** (first token in each edit) as critical anchors for guiding subsequent corrections and reducing error accumulation.
- Developed an annotation-free preference data construction method by sampling negative sentences with large edit distances, enabling cost-effective training.
- Achieved forefront single-model performance on English and Chinese GEC benchmarks, surpassing supervised fine-tuning and Direct Preference Optimization, while mitigating overcorrection and improving robustness.

Research on Chinese Text Correction System Based on Deep Learning

Dec 2022 – May 2023

Outstanding Undergraduate Thesis (top 10%), Supervised by Prof. Binbin Yong at Lanzhou University

- Designed a comprehensive **Chinese text correction system** covering spelling, grammar, and semantic errors.
- Proposed **AutoCscError**, a novel pseudo-data generation and augmentation method that predicts likely human spelling errors using shape/phonetic similarity enhanced by IDS encoding, significantly improving zero-shot and fine-tuned performance across multiple benchmarks by producing more realistic pseudo-data.

A Novel Dynamic Interpolation Method based on Temporal and Spatial Correlations

May 2021 – Oct 2021

Research Team Member, Supervised by Prof. Zhili Zhao at Lanzhou University

- Proposed a **Dynamic Spatiotemporal Interpolation** method for environmental and meteorological monitoring, which adaptively weights temporal and spatial correlations for more accurate missing data recovery.

SERVICES

- **Reviewer:** NeurIPS2026, EMNLP2026;
- **Teaching Assistant:** Artificial Neural Networks in Sun Yat-sen University;

TECHNICAL SKILLS

Languages	Python, C++, Java, Bash, LaTeX, SQL
Frameworks & Libraries	Transformers, Ray, VeRL, LLaMA-Factory, Datasets, Accelerate, DeepSpeed, vLLM
Tools & Platforms	Git, Docker, Weights & Biases, SwanLab, TensorBoard, Slurm, CUDA